

KrishiMitra — Backend Database & SIH Q&A; Guide

Project: KrishiMitra | **Stack:** Node.js + PostgreSQL (Supabase) + Prisma ORM | **Purpose:** SIH Presentation & Technical Q&A; Defense

1. High-Level Database Architecture & Stack Choice

KrishiMitra ka database architecture 3 main layers par built hai. System ka primary goal hai **100% ACID Compliance**, **Zero Data Loss**, aur **High Concurrency** during Mandi peak hours.

Component	Technology Choice	Architectural Rationale
Database Engine	PostgreSQL (Supabase)	Strict relational integrity, foreign key constraints & financial transaction safety.
ORM Layer	Prisma ORM v5	Type-safe database queries, auto-generated TypeScript/JS client, and schema migration management.
Backend API	Node.js & Express (Render)	Asynchronous REST API backend communicating with DB via connection pooling.
Frontend App	React 19 + Vite (Vercel)	Dynamic single-page client consuming backend APIs.

2. Database Connection Lifecycle & Configuration

Backend Express server aur Supabase PostgreSQL DB ke beech connection setup dual connection strings se managed hai:

Connection Mode	Target URL & Port	Usage & Purpose
Pooled Connection	DATABASE_URL (PgBouncer - Port 6543)	Runtime transaction-level pooling so concurrent requests do not exhaust connection limits.
Direct Connection	DIRECT_URL (Direct Socket - Port 5432)	Reserved exclusively for Prisma DDL migrations and schema updates.

3. Database Schema Breakdown (22 Models in 5 Modules)

Data model prisma/schema.prisma me defined 22 tables 5 core logical modules me divided hain:

Module	Models / Tables	Key Responsibilities & Schema Features
1. User & Auth	User, FarmerProfile, StaffProfile, FarmerVerificationHistory	User roles (FARMER, STAFF, ADMIN), profile verification status, masked Aadhaar & Bank hashes.
2. Infrastructure	ProcurementCentre, Counter, StaffAssignment, SlotConfig, SlotAllocation	Mandi location coordinates, daily procurement capacity, operational counters, and slot time bounds.

3. Crop & Pricing	Crop, CropGrade, CropPrice, ProcurementSeason, FarmerCropRegistration	Government MSP price rates, quality grades, procurement seasons, and land crop registration.
4. Queue & Slot	ProcurementBooking, QueueToken, TokenStatusHistory	Slot booking engine, real-time counter tokens, and compound unique key constraints for zero double-booking.
5. Fulfillment & Audit	ProcurementTransaction, WeighingRecord, QualityInspection, Payment, AuditLog, TrustScoreHistory	Gross/tare/net crop weighment, moisture quality tests, automated payments, and complete system audit logs.

4. Word-for-Word Speaking Script for SIH Presentation

Aap is script ko SIH Judges ke samne directly deliver kar sakte hain (3 Minutes Presentation):

Part 1: Intro & Core Philosophy (30 Seconds)

"Good morning / afternoon Respected Judges! Today I will present the core backbone of KrishiMitra — our Database Architecture and Data Lifecycle.

Jab hum Mandi Procurement aur Farmer Payments jaisa platform design karte hain, humara primary goal hota hai: **Zero Data Loss, High Security, aur 100% Transactional Accuracy**. Isi liye humne PostgreSQL on Supabase choose kiya. NoSQL ke bajaye Relational DB select karne ki main reason hai strict ACID Compliance, taaki kisi bhi payment ya crop transaction me data mismatch na ho.

Backend me hum Node.js & Express REST APIs ke sath Prisma ORM use kar rahe hain jo type-safety aur query performance optimization provide karta hai."

Part 2: End-to-End Data Flow (1.5 Minutes)

"Ab main aapko ek Farmer ki end-to-end journey me backend data flow samjhata hoon:

1. Secure Onboarding: Jab kisan register karta hai, details FarmerProfile table me save hoti hain. Data privacy ke liye hum Plain-Text Aadhaar ya Bank Account save nahi karte — hum Masked string (XXXX-XXXX-1234) ke sath SHA-256 Hashes store karte hain.

2. Smart Slot & Queue Token: Kisan Mandi slot book karta hai toh ProcurementBooking record create hota hai aur ek unique QueueToken issue hota hai. Overbooking aur race condition rokne ke liye humne schema level par Compound Unique Indexes ([centreId, date, slotTime, tokenNumber]) lagaye hain.

3. Mandi Processing: Mandi arrival par QualityInspection table update hoti hai (moisture % & grade) aur WeighingRecord me gross vs tare weight calculate karke net weight save hota hai.

4. Automated Payment & Audit: Quality pass hone par ProcurementTransaction create hota hai, net weight MSP rates se multiply hota hai, aur Payment status PENDING se SUCCESS update ho jata hai. Saare steps AuditLog table me capture hote hain fraud prevention ke liye."

Part 3: Engineering Highlights & Wrap-up (1 Minute)

"Judges, humare DB design me 3 critical engineering highlights hain:

- **First — Interactive Transactions (prisma.\$transaction):** Multi-table operations atomic transactions me hoti hain. Agar payment step fail hota hai, toh saare steps automatically Rollback ho jate hain.
- **Second — B-Tree Indexing:** High traffic filters like district, mobile, status, aur date par indexes hain jo $O(\log N)$ fast lookups guarantee karte hain.
- **Third — Connection Pooling:** Production me Supabase PgBouncer connection pooler active hai jo thousand concurrent requests smoothly handle karta hai.

In short, KrishiMitra ka database Security, Crowd Management, aur Automated Payment Verification ko backend level par enforce karta hai. Thank you!"

5. Top 10 Technical Q&A; with SIH Judges

In Q&A; pairs ko acche se tayyar karein for technical defense during judging rounds:

Q1: High Concurrency par slot double-booking kaise rokte ho?

Humne 2-layer defense mechanism use kiya hai: (1) Schema level par Compound Unique Index `@@unique([centreId, date, slotTime, tokenNumber])`, jo DB level par duplicate entries reject karta hai. (2) Prisma Interactive Transaction (`$transaction`) jahan slot check, capacity update, aur booking atomic block me execution hoti hai.

Code Reference: [schema.prisma:L319](#) | [booking.controller.js:L125](#)

Q2: Farmer Aadhaar aur Bank details leak hone se kaise protected hain?

Plain-text Aadhaar ya Bank AC details store nahi hoti. Display ke liye Masked Strings (`aadhaarMasked`, `accountNumberMasked`) store hote hain. Identity lookup aur uniqueness verify karne ke liye SHA-256 Hashes (`aadhaarHash`, `accountNumberHash`) rely kiye jaate hain jo 100% irreversible hain.

Code Reference: [schema.prisma:L92-L105](#)

Q3: MongoDB (NoSQL) ke bajaye PostgreSQL (SQL) kyun choose kiya?

NoSQL eventual consistency par work karta hai jo financial platforms ke liye risky hai. KrishiMitra me crop weighment, MSP calculations aur Direct Bank Payments involve hain. Strict ACID compliance, foreign key constraints, aur zero financial mismatch ke liye PostgreSQL mandatory choice thi.

Code Reference: [Architectural Decision — Financial Integrity](#)

Q4: Payment creation ke waqt server crash ho jaye toh kya data half-save hoga?

Nahi! All multi-table updates are wrapped inside Prisma Interactive Transactions (`prisma.$transaction`). Agar step 3 (Payment) fail ho jata hai, toh step 1 (Weighing) aur step 2 (Quality) automatically Rollback ho jate hain. DB state completely clean aur unchanged rehti hai.

Code Reference: [procurement.controller.js:L46](#)

Q5: Staff member crop weight ya grade me fraud kare toh kaise catch karoge?

System me Complete Auditability enforcement hai: `WeighingRecord` me `staff operatorId`, `QualityInspection` me `inspectorId` linked hoti hai. Global `AuditLog` table me exact user ID, entity name, entity ID, aur timestamp automatically populate hota hai.

Code Reference: [schema.prisma:L385, L423, L462](#)

Q6: Tens of thousands of records par queries slow nahi hongi?

Queries $O(\log N)$ execution complexity me perform karti hain kyunki humne target B-Tree Indexes (@@index) configure kiye hain on high-frequency query columns like district, mobile, status, centreId, aur effectiveDate.

Code Reference: [schema.prisma:L116-L118, L320-L325](#)

Q7: Cloud backend hits se Connection Limit Exhaust ('Too Many Connections') kaise prevent hota hai?

Humne Two-URL Connection Architecture setup kiya hai. Application runtime DATABASE_URL connects through Supabase PgBouncer (Port 6543) for transaction connection pooling, while DIRECT_URL (Port 5432) is reserved exclusively for Prisma schema migrations.

Code Reference: [schema.prisma:L3-L4](#)

Q8: Farmer account delete hone par historical transaction data ka kya hoga?

FarmerProfile level par onDelete: Cascade enabled hai for profile cleanup, but financial records (ProcurementTransaction, Payment) auditing compliance ke liye retained rehte hain. Account suspension status: SUSPENDED Enum through handle hota hai.

Code Reference: [schema.prisma:L20, L87](#)

Q9: Counter Queue System live updates DB se real-time kaise synced rehta hai?

QueueToken model me status (WAITING, CALLED, PROCESSING, COMPLETED) aur timestamps track hote hain. Jab staff token call karta hai, status update DB me write hote hi Express backend real-time state push karta hai.

Code Reference: [schema.prisma:L327-L360](#)

Q10: Farmer ka Trust Score calculation DB me kaise log hota hai?

FarmerProfile me trustScore: 100.0 initial baseline hota hai. Timely mandi arrival ya standard quality crop landing par TrustScoreHistory table me event and points log hote hain. Absence ya violation par negative points deduct hoke score update hota hai.

Code Reference: [schema.prisma:L107, L283-L291](#)